

OS LAB WEEK-5

Proportional Share Scheduling

```
#include<stdio.h>

#include <stdlib.h>

#include <time.h>

typedef struct
{
    char name[5]; int tickets; }
    Process;

int main() {
    int n, total_tickets = 0; float total_T =0.0;
    printf("Enter the number of Processes: ");
    scanf("%d", &n);

    Process p[n];
    srand(time(NULL));

    for (int i = 0; i < n; i++)
    { printf("\nProcess %d:\n", i + 1);
      sprintf(p[i].name, "P%d", i + 1);
      printf("Tickets: ");
      scanf("%d", &p[i].tickets);
      total_tickets += p[i].tickets;
      total_T +=p[i].tickets;
    }

    printf("\n--- Proportional Share Scheduling ---\n");
    printf("Enter the Time Period for scheduling: ");
    int m;

    scanf("%d",&m);
    for (int i = 0; i < m; i++)
    {
```

```

int winning_ticket = rand() % total_tickets + 1;

int accumulated_tickets = 0;

int winner_index;

for (int j = 0; j < n; j++)
{
    accumulated_tickets += p[j].tickets;
    if (winning_ticket <= accumulated_tickets)
    {
        winner_index = j; break;
    }
}

printf("Tickets picked: %d, Winner: %s\n", winning_ticket,
p[winner_index].name);
}

for (int i = 0; i < n; i++)
{
    printf("\nThe Process: %s gets %0.2f%% of Processor Time.\n", p[i].name,
((p[i].tickets / total_T) * 100));

} return 0; }

```

Enter the number of Processes: 3

Process 1:

Tickets: 10

Process 2:

Tickets: 20

Process 3:

Tickets: 30

--- Proportional Share Scheduling ---

Enter the Time Period for scheduling: 5

Tickets picked: 21, Winner: P2

Tickets picked: 41, Winner: P3

Tickets picked: 7, Winner: P1

Tickets picked: 60, Winner: P3

Tickets picked: 48, Winner: P3

The Process: P1 gets 16.67% of Processor Time.

The Process: P2 gets 33.33% of Processor Time.

The Process: P3 gets 50.00% of Processor Time.

Process returned 0 (0x0) execution time : 23.494 s

Press any key to continue.

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